

Nitesh Shantha Kumar

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EXPERIENCE

Nordstrom

Seattle, WA · Aug 2021 – Present

Software Engineer — Pricing and Promotions Platform

- Technical lead for a clearance and promotions platform with two ingestion paths, a synchronous REST API and an asynchronous Kafka consumer, plus a business-user application; decomposed delivery into dependency-mapped worktracks so four engineers could build in parallel, owning design review and integration across them.
- Migrated clearance and end-of-lifecycle event publishing off a legacy Oracle and Airflow pipeline onto Kafka using a transactional outbox with confirmed-flush acknowledgement, publishing to existing topics with identical schemas so downstream consumers required zero configuration changes. Cutover gated on a production shadow-mode parity check behind split feature flags, with tiered rollback covering named failure scenarios.
- Designed the replacement of ~28,000 blocking per-clearance product-service calls/day with a log-compacted Kafka topic consumed into a local read store, keeping the upstream service as a permanent fallback on miss. Baselined production latency and error rate before designing (32ms average, p99 199ms, 0.05% errors across 197,579 calls). Authored three architecture decision records, each naming the rejected alternative and the constraint behind it.
- Chose asynchronous batch processing over synchronous downstream calls, then rate-limited to 2K clearances/min against a 10-minute SLA to protect upstream inventory, pricing and conflict services despite 5x headroom; load tested to 2.5x the production limit.
- Root-caused a processing backlog of ~20,000 clearances within 1h18m: the job loaded the entire queue per cycle against an upstream service resolving ~7 records/sec with per-request batching unavailable, giving a hard ceiling near 350 rows per 50-second cycle. Shipped bounded batching as containment, then per-call timeouts and commit-and-continue resilience as the structural fix; published an RCA with five causal chains and 14 prioritised follow-ups.
- Cut PostgreSQL CPU from 100% to under 10% within 24 hours of diagnosis: traced 5.6B sequential row reads with zero index scans to a COALESCE blocking index use on a view's join keys, proving the keys were never null and row output identical before proposing (EXPLAIN cost 82,460 → 13; rows scanned 7.5M → 3). Removed the need for a proposed user-facing restriction.
- Redesigned promotion redemption limit enforcement as a single atomic database operation combining validation and counter increment, eliminating race-condition limit violations; P99 redemption latency 1,000ms → 10ms with support for 10x concurrency.
- Built and published an agentic incident-investigation toolkit as an installable plugin: 10 skills and 7 subagents automating triage across Slack, New Relic, GitHub, Confluence, ServiceNow and Jira, with 36 structured root-cause documents; extended by another engineer in a 1,557-line pull request.
- Built the platform alerting programme after two incidents where monitoring appeared healthy but could not fire: 11 category pages pairing every alert condition to concrete triage steps, with 7 of 9 proposed alerts now live in production.

Ak Aerotek

Bengaluru, India · May 2016 – May 2019

Software Engineer

- Verification and validation of safety-critical avionics software to DO-178B/C, authoring Ada test harnesses and executing them against target hardware.
- Led a team of 6 through code and audit reviews, reducing the delivery defect rate by 90%.

EDUCATION

M.S., Engineering Sciences (Robotics) — University at Buffalo, NY

2019 – 2021

Graduate Research Assistant, autonomous-vehicle perception and sensor fusion (LANDR Lab)

B.E., Electronics & Communication — Visvesvaraya Technological University, India

2011 – 2015

TECHNICAL SKILLS

Languages	Java, Python, SQL, C++
Distributed systems	Apache Kafka (producers, consumers, topic ACLs, Avro, log compaction, Confluent Cloud), transactional outbox and event-driven architecture, RESTful API design, service-oriented architecture, large-scale distributed systems, concurrency and multi-threading, fault tolerance
Data	PostgreSQL, Oracle, DynamoDB, Neo4j, Redis / Valkey (ElastiCache)
Cloud & operations	AWS (EKS, S3, RDS, Aurora), Kubernetes, Docker, New Relic, Splunk, JMeter, Flyway, incident response and on-call